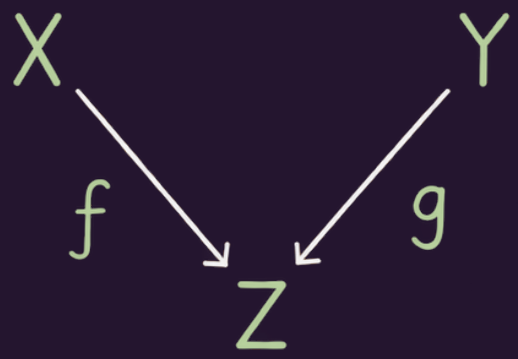
Factoring in (the other) Pre-Hilbert *-Categories

A generalisation of Douglas' lemma

Matthew Di Meglio

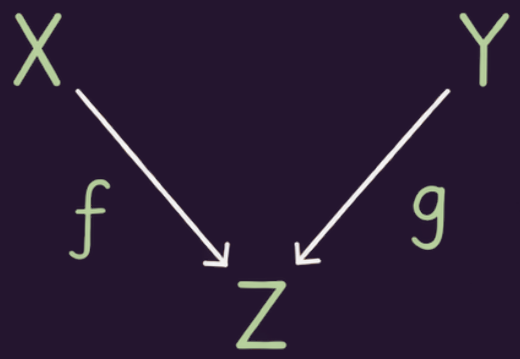
Category Theory Octoberfest 2026

Douglas' Lemma for Hilbert Spaces



$\operatorname{Ran} f \leqslant \operatorname{Ran} g$ (range inclusion)

Douglas' Lemma for Hilbert Spaces

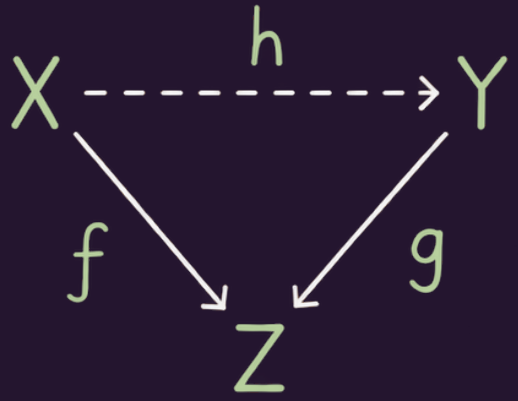


$$\operatorname{Ran} f \leqslant \operatorname{Ran} g \quad \text{(range inclusion)}$$

$$\Updownarrow$$

$$ff^* \leqslant \lambda^2 gg^* \quad \text{for some } \lambda \geqslant 0 \quad \text{(majorisation)}$$

Douglas' Lemma for Hilbert Spaces



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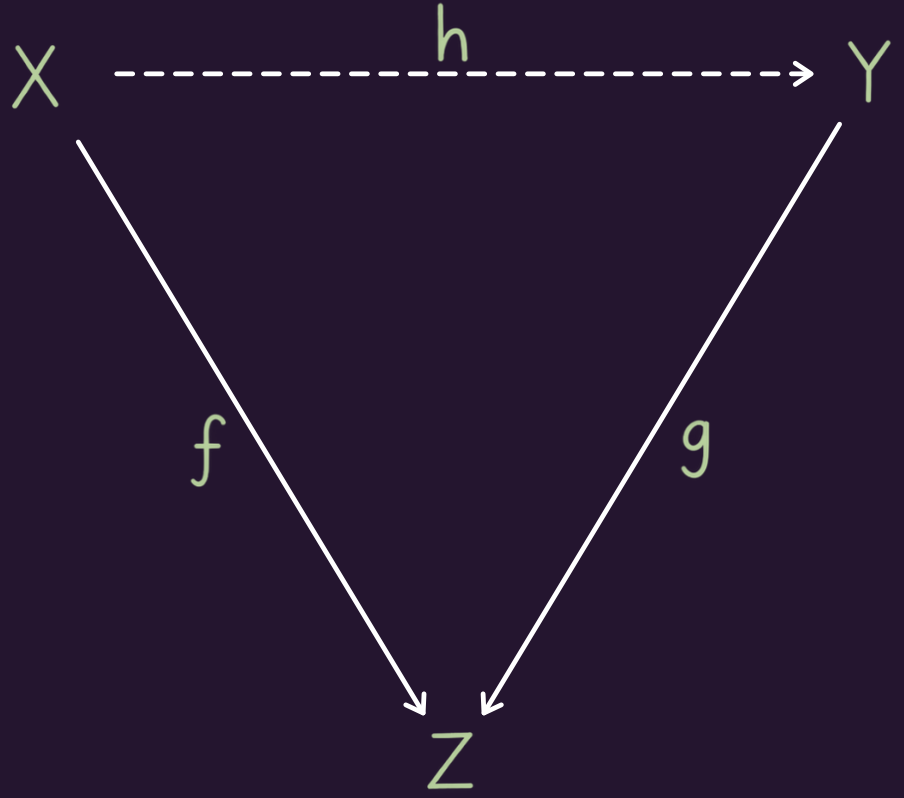
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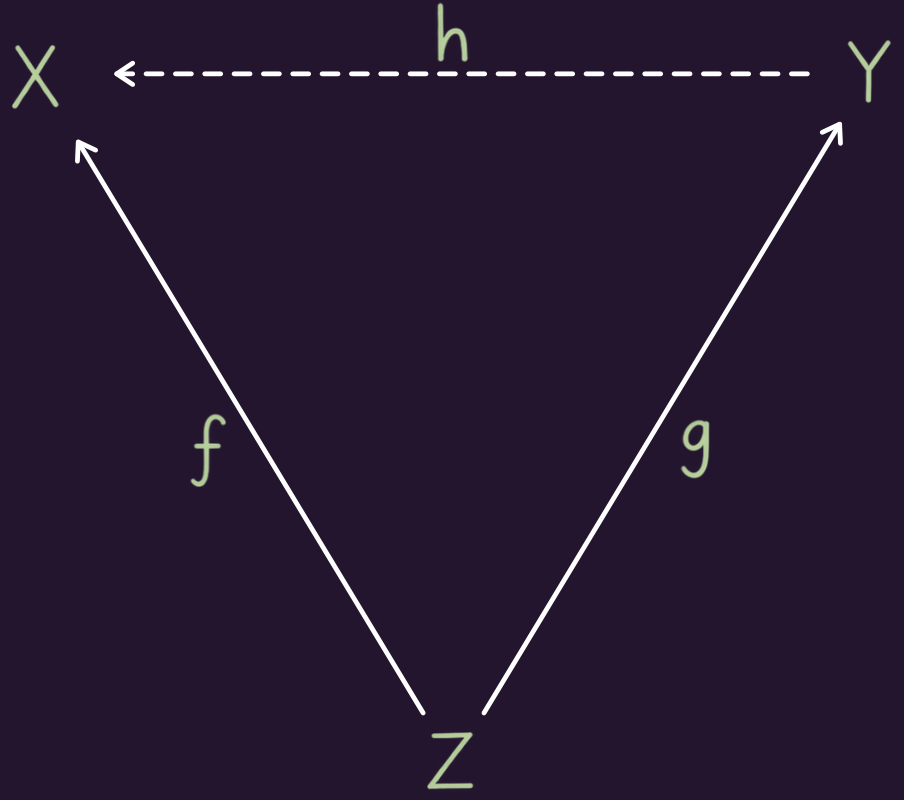
$$f = gh \quad \text{for some bounded } h \quad \text{(factorisation)}$$

The Analytic Proof



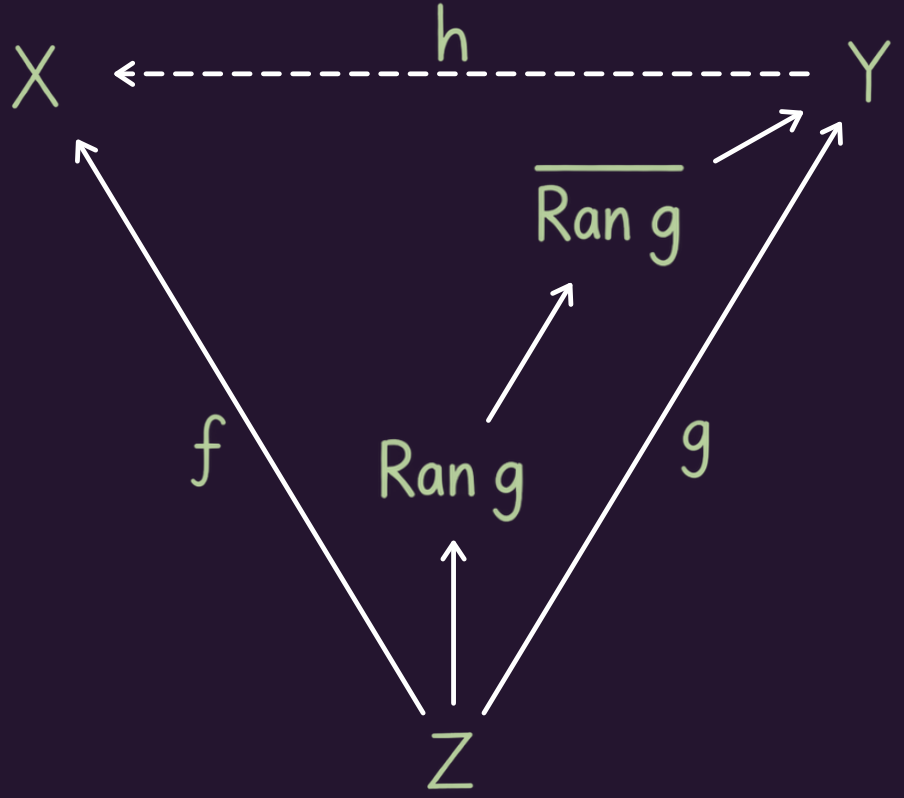
$$ff^* \leqslant \lambda^2 gg^* \Rightarrow f = gh$$

The Analytic Proof



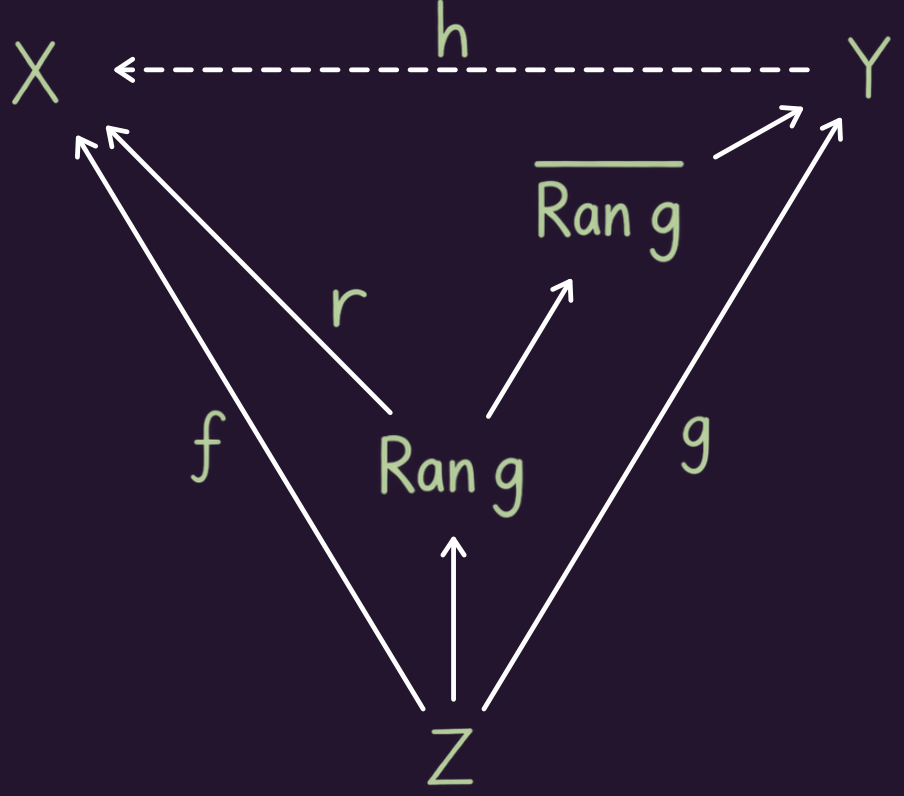
$$f^* f \leqslant \lambda^2 g^* g \Rightarrow f = h g$$

The Analytic Proof



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The Analytic Proof



$$f^*f \leqslant \lambda^2 g^*g \Rightarrow f = hg$$

$$\|fz\| \leqslant \lambda \|gz\|$$

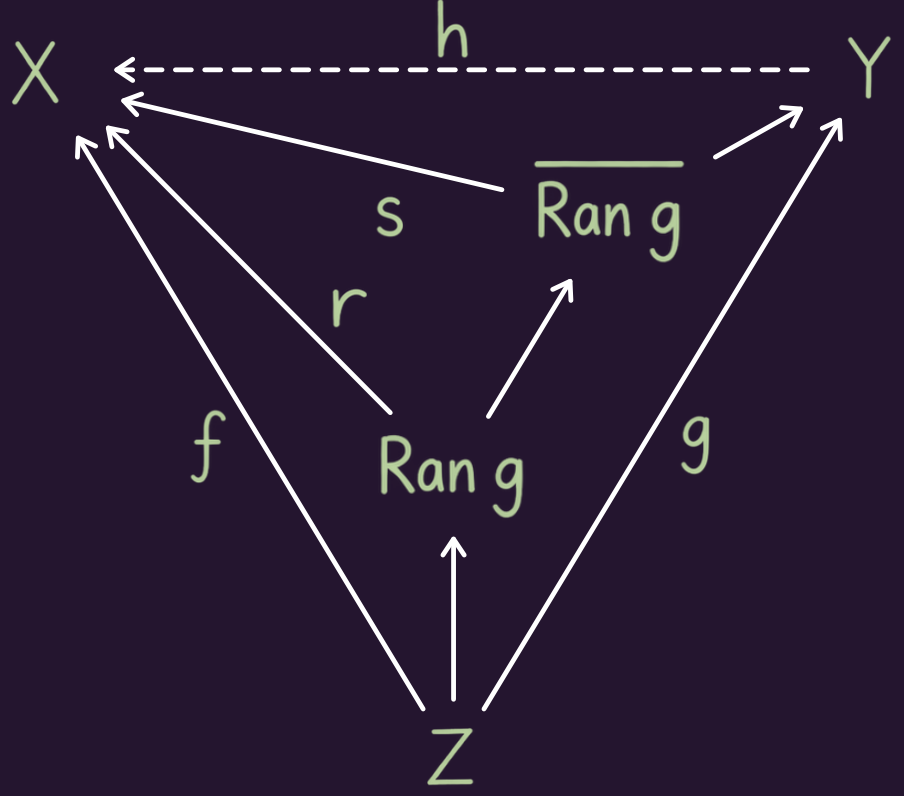
$$r(gz) = fz$$

$$gz = 0 \Rightarrow fz = 0$$

well-defined and bounded

$$\|r\| \leqslant \lambda$$

The Analytic Proof



$$f^*f \leqslant \lambda^2 g^*g \;\Rightarrow\; f = h\,g$$

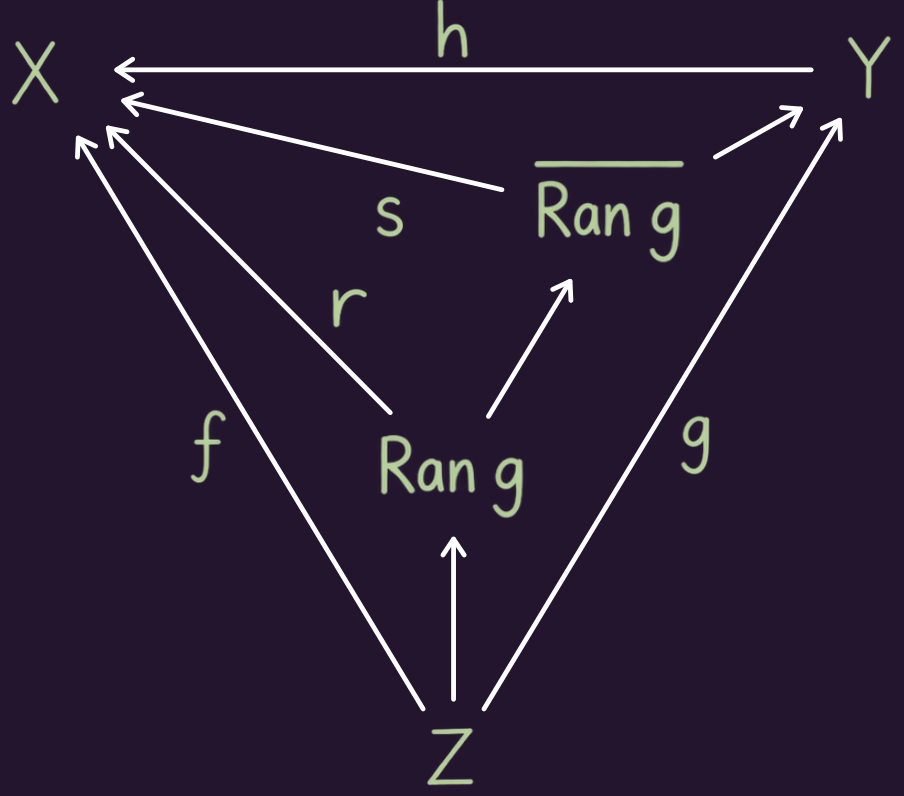
continuity + completeness

$$y_n \in \text{Ran } g\,, \quad y_n \longrightarrow y$$

$$s\,y = \lim_{n\to\infty} r\,y_n$$

$$\|s\| \leqslant \lambda$$

The Analytic Proof



$$f^* f \leq \lambda^2 g^* g \Rightarrow f = h g$$

$$Y = \overline{\text{Ran}\, g} \oplus (\text{Ran}\, g)^{\perp}$$

$$h = [s \;\; 0]$$

$$\|h\| \leq \lambda$$

Pre-Hilbert $*$-categories

A $*$-category $\mathbf{C}$ is pre-Hilbert if:

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Hilbert spaces and bounded linear maps

$\{0\}$

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Hilbert spaces and bounded linear maps

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$X \oplus Y = \{(x, y) \mid x \in X,\ y \in Y\}$

$\langle (x, y), (x', y') \rangle = \langle x, x' \rangle + \langle y, y' \rangle$

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$\operatorname{Ker} f = \{x \in X \mid fx = 0\}$

with the inherited inner product

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Axioms are algebraic and finitary

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zero morphisms

Axioms are algebraic and finitary

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A $*$-category $\mathbf{C}$ is pre-Hilbert if:

- it has a zero object — zero morphisms
- it has binary orthonormal biproducts — addition, matrix calculus
- it has isometric kernels
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Axioms are algebraic and finitary

Pre-Hilbert $*$-categories

A $*$-category $\mathbf{C}$ is pre-Hilbert if:

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- it has isometric kernels — orthogonal complements, closure
- every diagonal is a kernel

Axioms are algebraic and finitary

Pre-Hilbert *-categories

A *-category **C** is pre-Hilbert if:

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- it has binary orthonormal biproducts — addition, matrix calculus
- it has isometric kernels — orthogonal complements, closure
- every diagonal is a kernel — subtraction

Axioms are algebraic and finitary

Examples

- Hilbert spaces and bounded linear maps

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- Hilbert spaces and bounded linear maps
- Unitary representations of a group

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- Hilbert spaces and bounded linear maps
- Unitary representations of a group
- Self-dual Hilbert modules over a complex W*-algebra

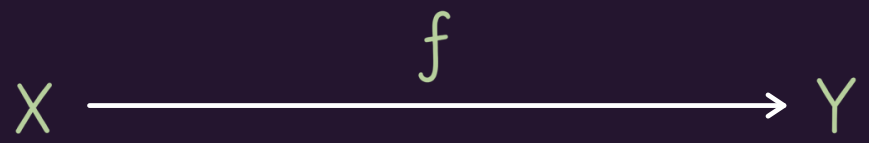
Ranges Categorically

poset of subobjects $\cong$ poset of semiclosed subspaces

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Ran f is smallest subobject of Y that f factors through



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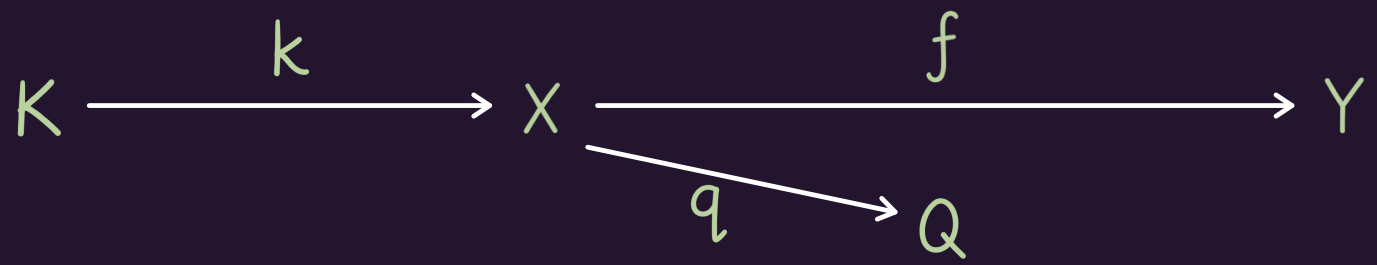
$[k] = \operatorname{Ker} f$

Every pre-Hilbert $*$-category is quasi-abelian

Ranges Categorically

poset of subobjects $\cong$ poset of semiclosed subspaces

Ran f is smallest subobject of Y that f factors through



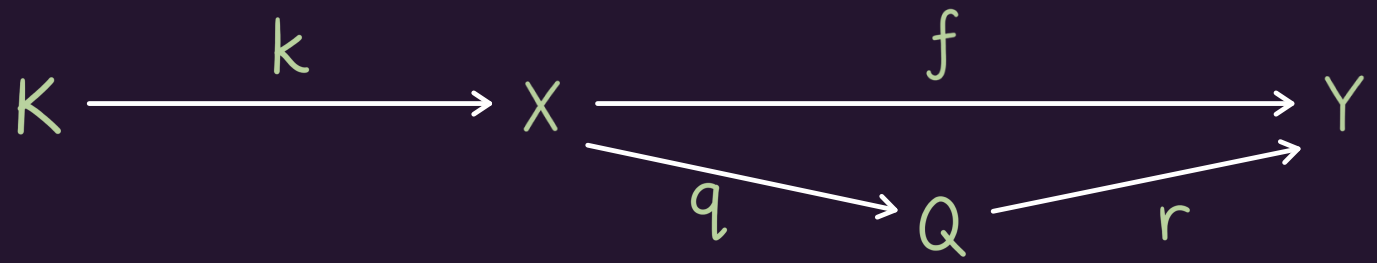
$[k] = \operatorname{Ker} f \qquad [q^*] = (\operatorname{Ker} f)^\perp$

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Ranges Categorically

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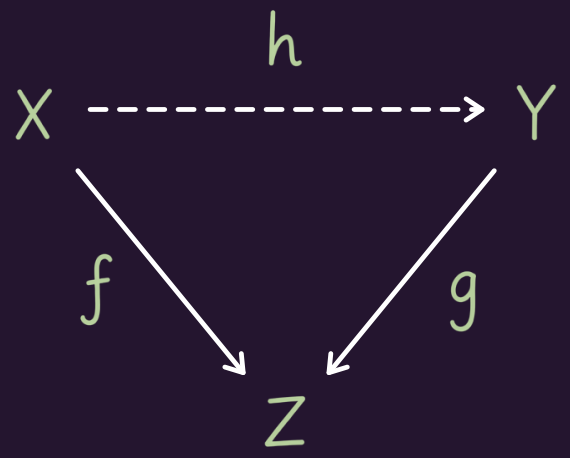
Ran f is smallest subobject of Y that f factors through



$[k] = \operatorname{Ker} f \qquad [q^*] = (\operatorname{Ker} f)^\perp \qquad [r] = \operatorname{Ran} f$

Every pre-Hilbert $*$-category is quasi-abelian

Douglas' Lemma Categorically

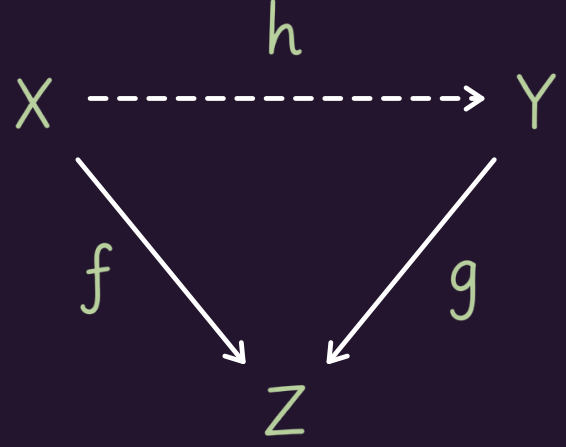


$$\mathrm{Ran}\ f \leqslant \mathrm{Ran}\ g$$

$$\Updownarrow$$

$$f = gh \quad \text{for some } h$$

Douglas' Lemma Categorically



$$\text{Ran}\, f \leqslant \text{Ran}\, g$$

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$f = gh$ for some h

For all $q \in \mathbb{Q}$

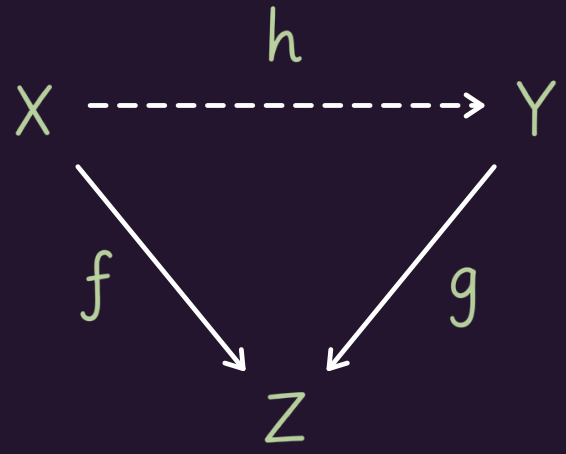
$$ff^* \leqslant q^2 gg^*$$

$$\Updownarrow$$

$$f = gh$$

for some h such that $hh^* \leqslant q^2$

Douglas' Lemma Categorically



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$\Updownarrow$

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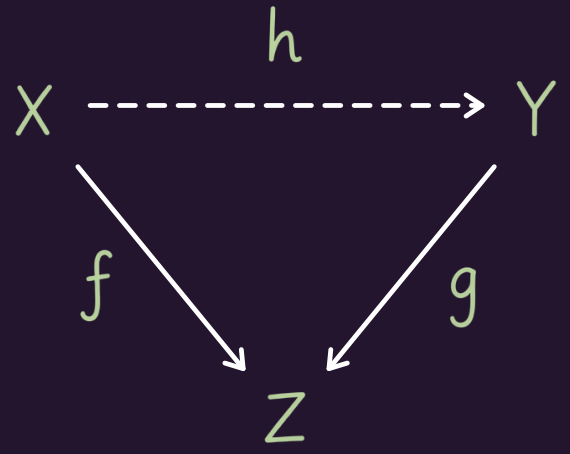
$f = gh$

for some h such that $hh^* \leqslant q^2$

$a \leqslant b \iff b - a = x^* x$

Rational scalars follow from the axioms.

Douglas' Lemma Categorically



$\mathrm{Ran}\ f \leqslant \mathrm{Ran}\ g$

$\Updownarrow$

$f = gh$ for some h

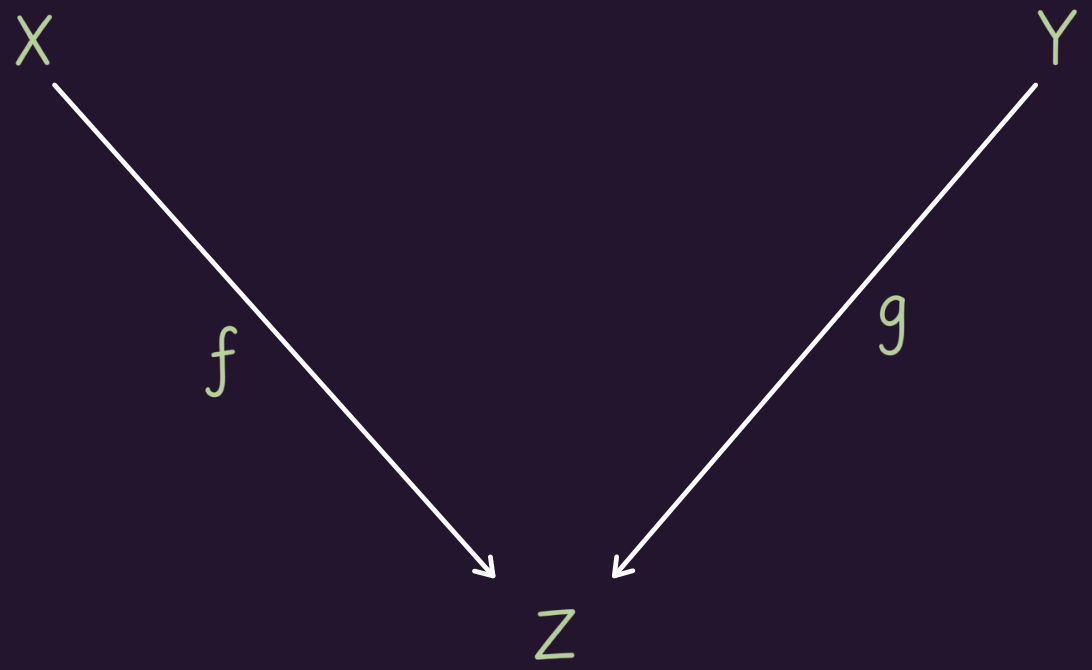
$ff^* = gg^*$

$\Updownarrow$

$f = gh$ and $g = fh^*$

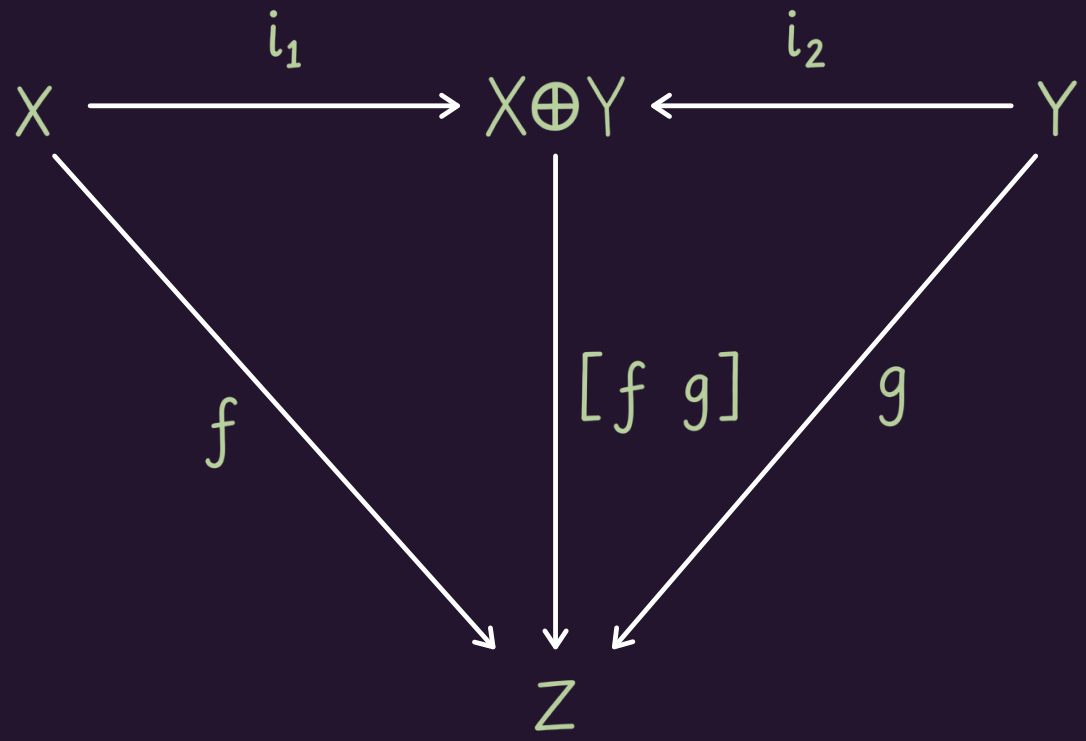
for some h such that $hh^*h = h$

The Algebraic Proof



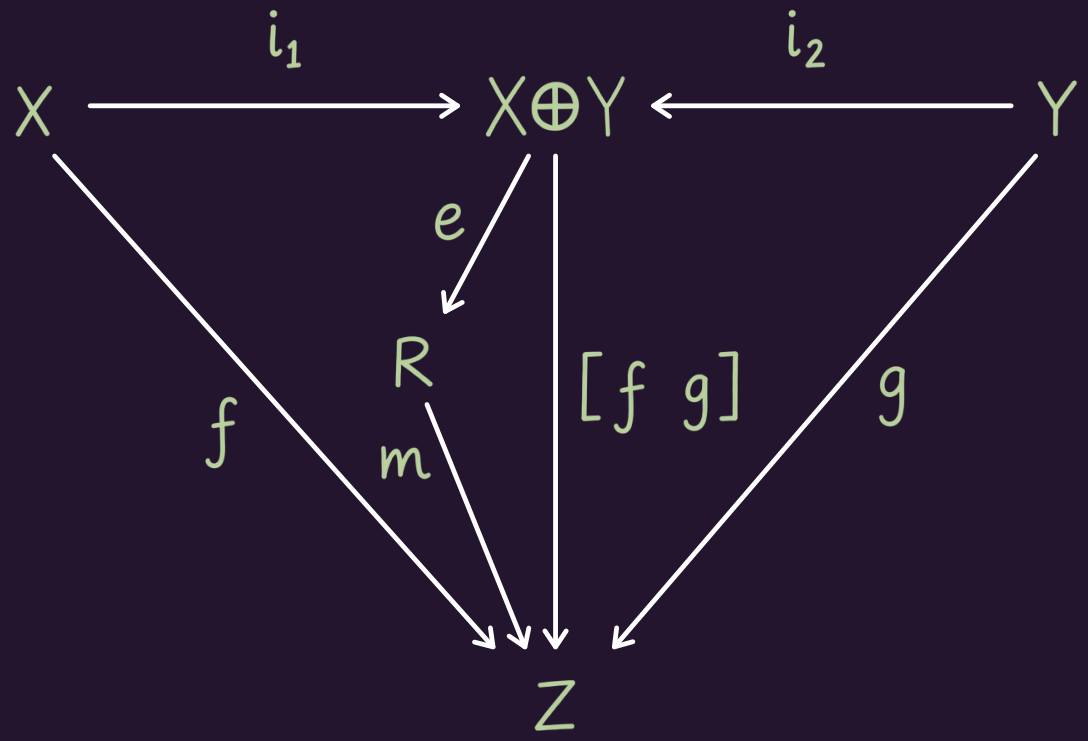
$$ff^* = gg^* \implies f = gh,\ g = fh^*,\ h = hh^*h$$

The Algebraic Proof



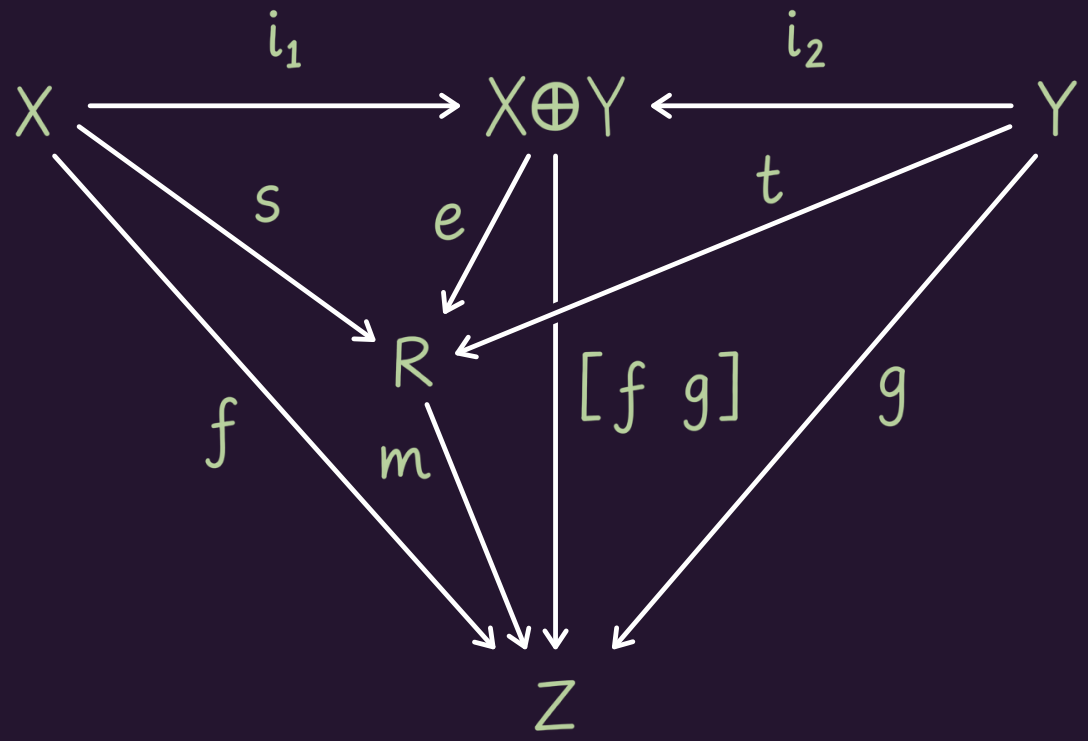
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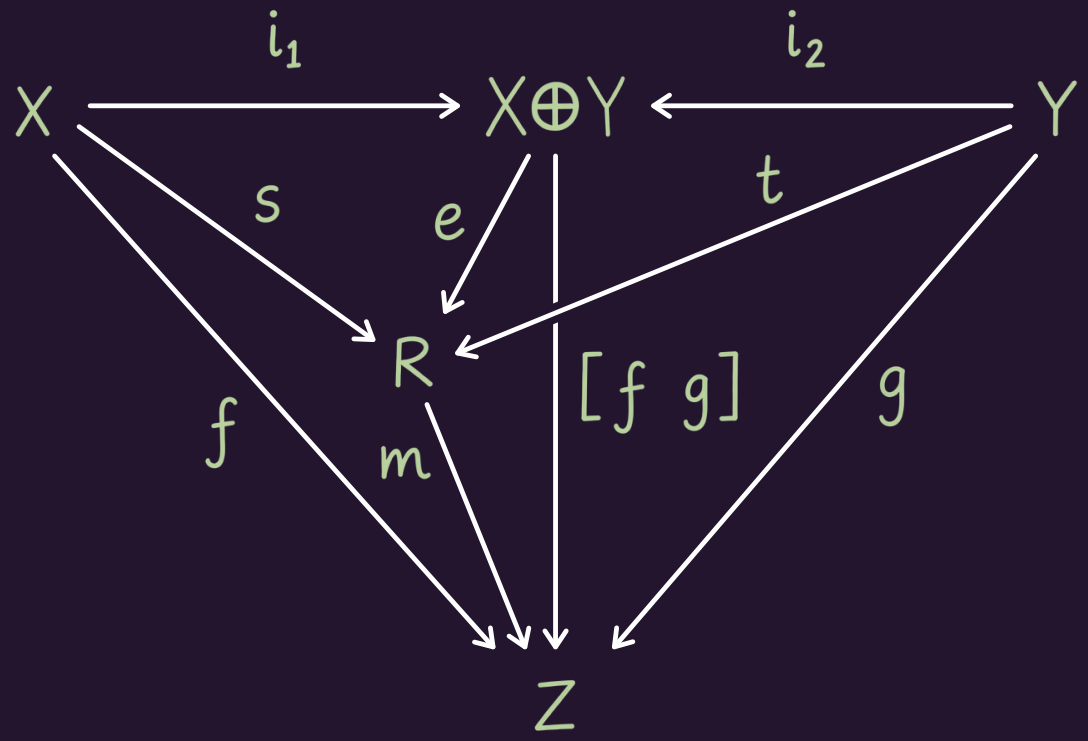
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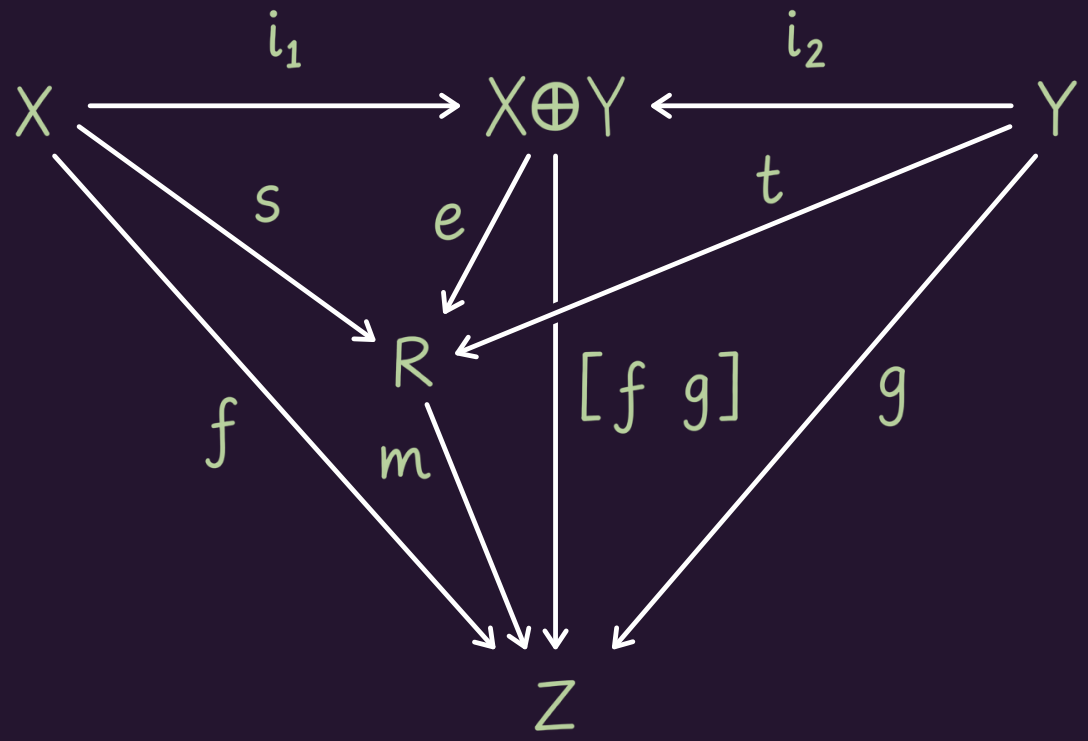
The Algebraic Proof



mss^*m^*

$ff^* = gg^* \Rightarrow f = gh,\ g = fh^*,\ h = hh^*h$

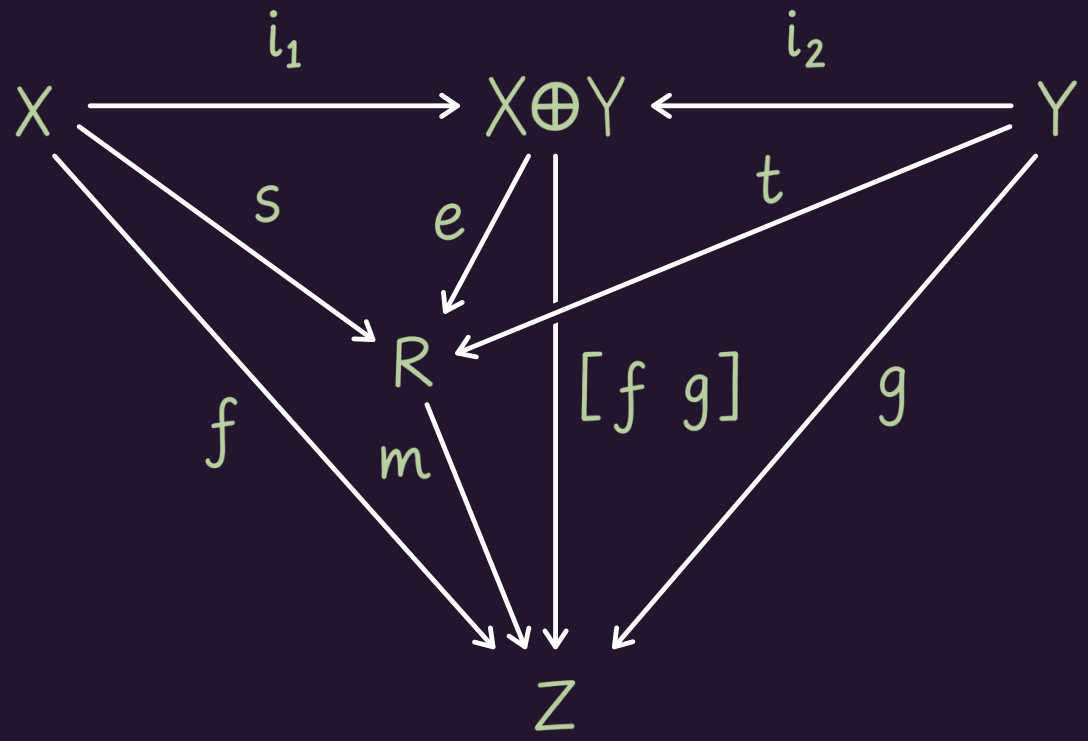
The Algebraic Proof



$mss^*m^* = ff^*$

$ff^* = gg^* \Rightarrow f = gh,\ g = fh^*,\ h = hh^*h$

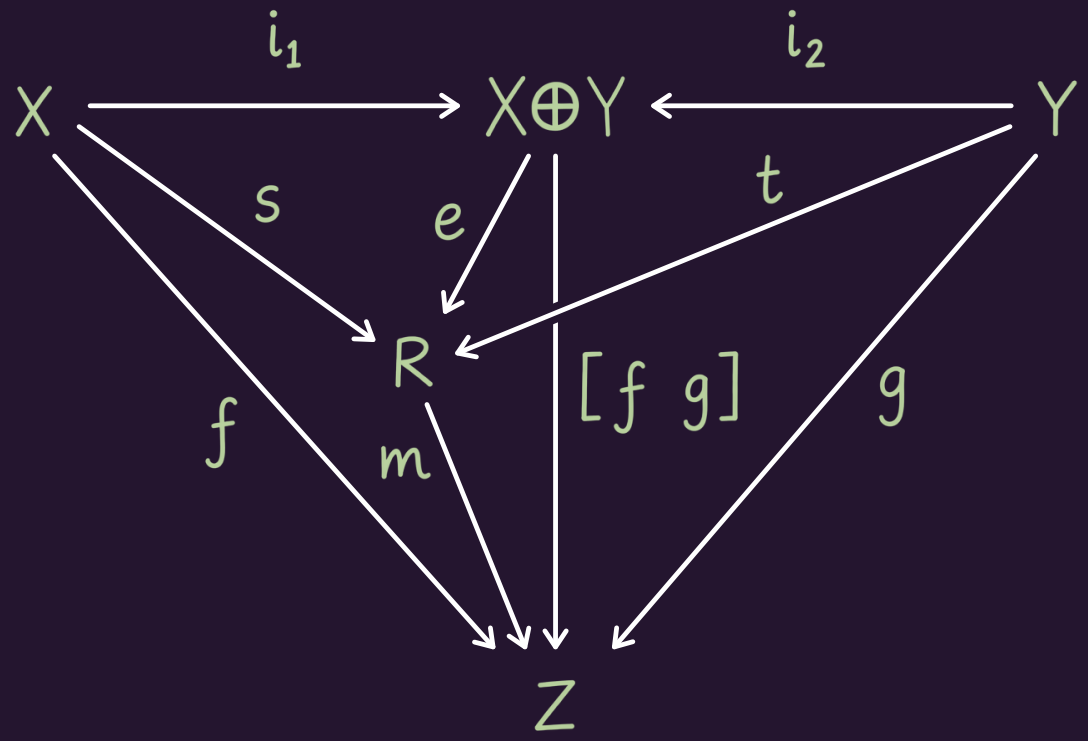
The Algebraic Proof



$mss^*m^* = ff^* = gg^*$

$ff^* = gg^* \Rightarrow f = gh,\ g = fh^*,\ h = hh^*h$

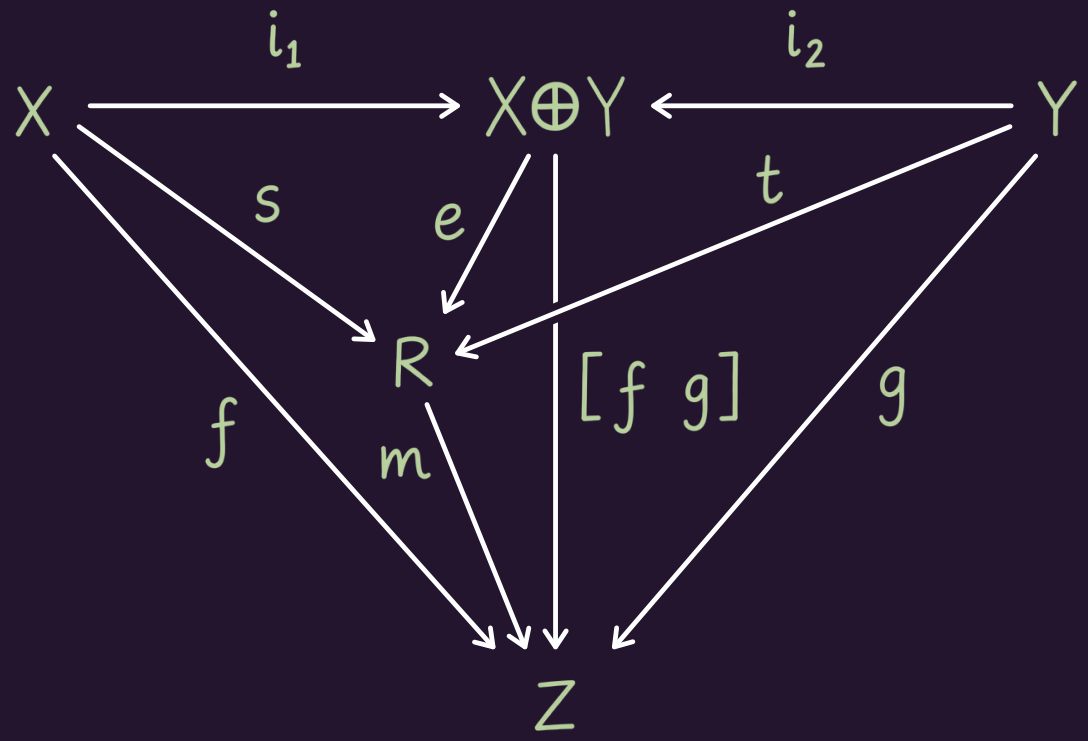
The Algebraic Proof



$ff^* = gg^* \Rightarrow f = gh,\ g = fh^*,\ h = hh^*h$

$mss^*m^* = ff^* = gg^* = mtt^*m^*$

The Algebraic Proof



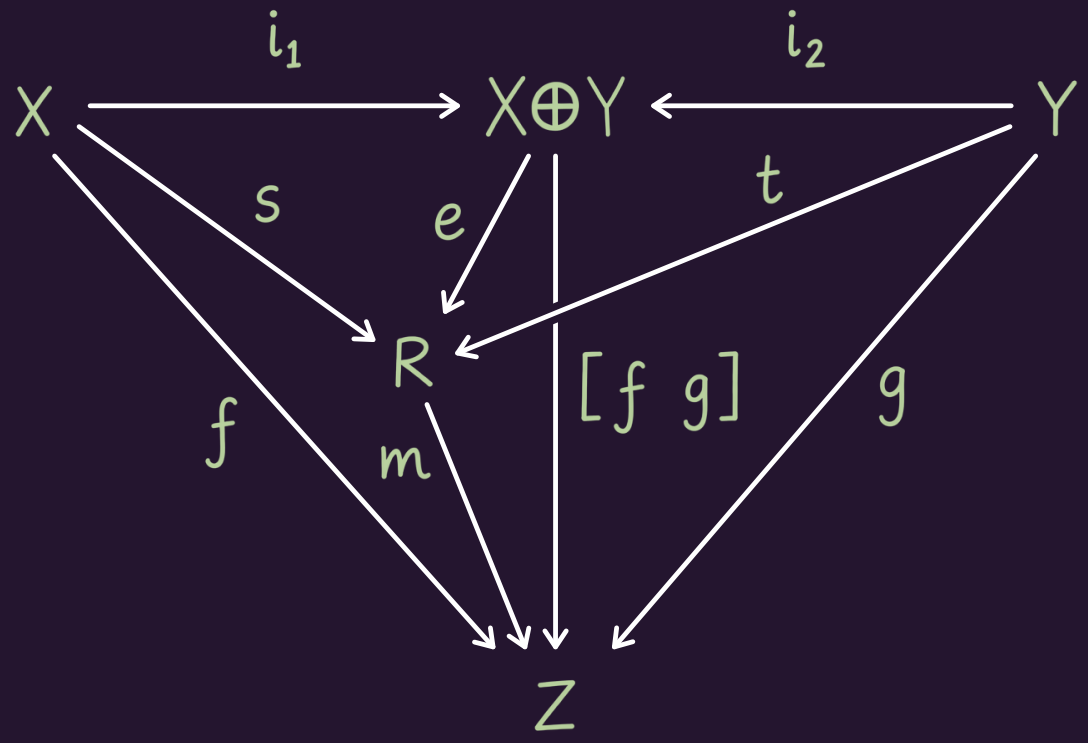
$ff^* = gg^* \Rightarrow f = gh,\ g = fh^*,\ h = hh^*h$

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$\Downarrow$

$ss^* = tt^*$

The Algebraic Proof



$ff^* = gg^* \Rightarrow f = gh,\ g = fh^*,\ h = hh^*h$

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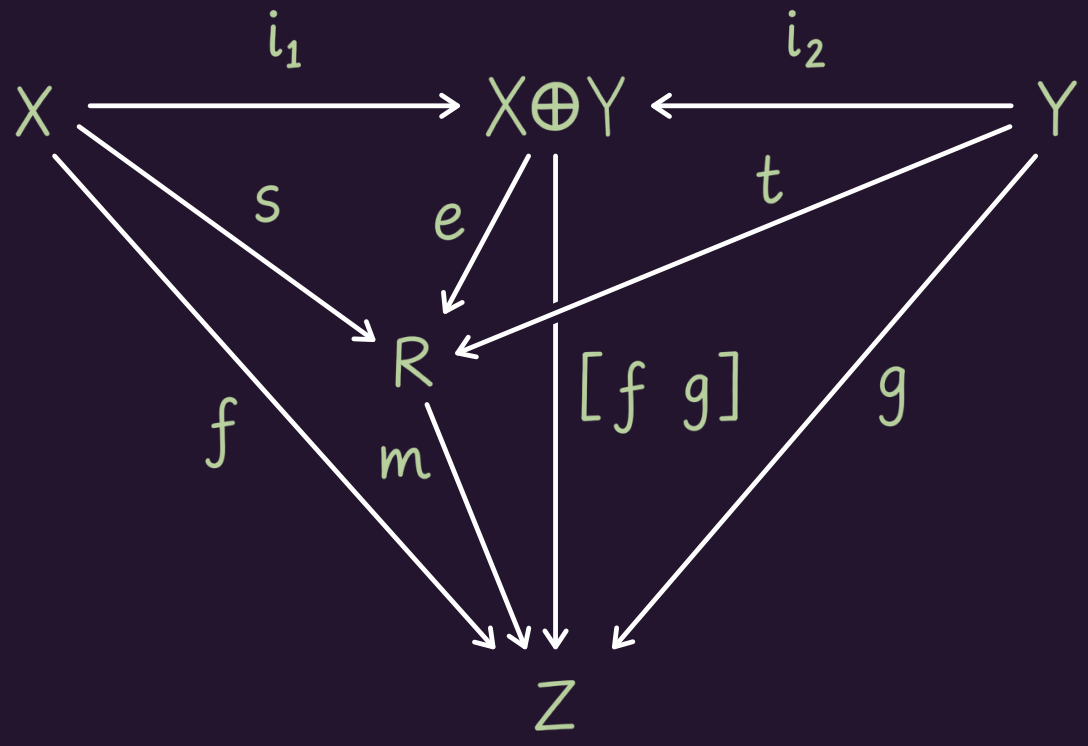
$\Downarrow$

$ss^* = tt^*$

$\Downarrow$

$1 = ee^*$

The Algebraic Proof



$ff^* = gg^* \Rightarrow f = gh,\ g = fh^*,\ h = hh^*h$

$$mss^*m^* = ff^* = gg^* = mtt^*m^*$$

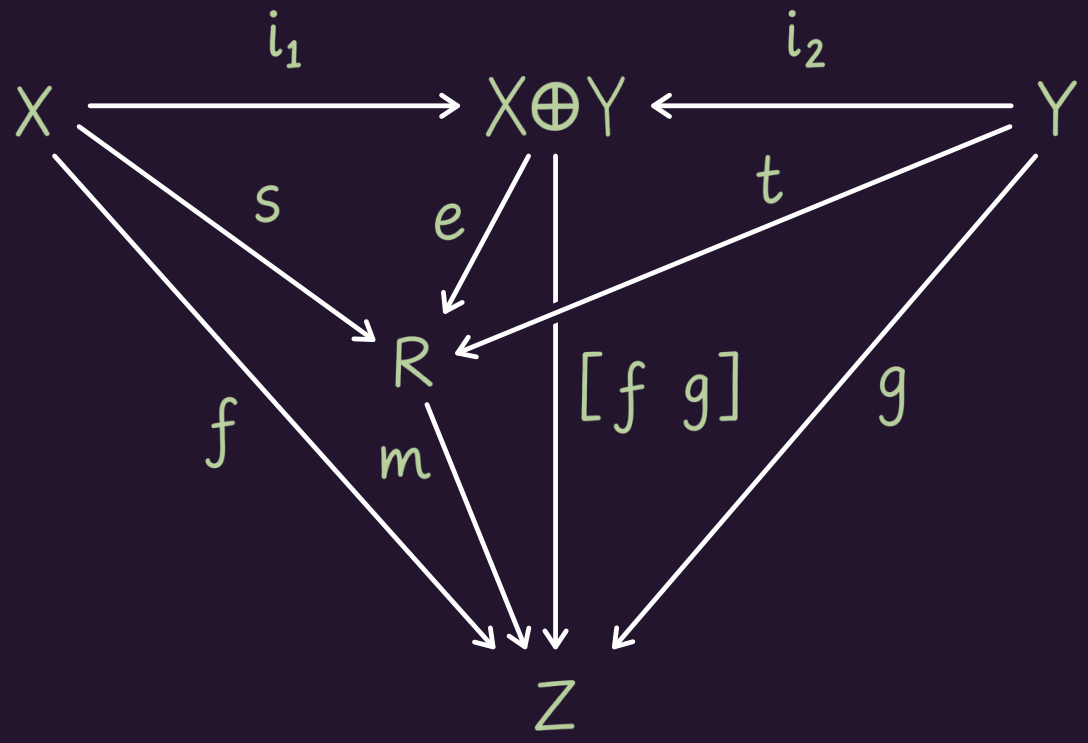
$$\Downarrow$$

$$ss^* = tt^*$$

$$\Downarrow$$

$$1 = ee^* = ss^* + tt^*$$

The Algebraic Proof



$ff^* = gg^* \Rightarrow f = gh,\ g = fh^*,\ h = hh^*h$

$$mss^*m^* = ff^* = gg^* = mtt^*m^*$$

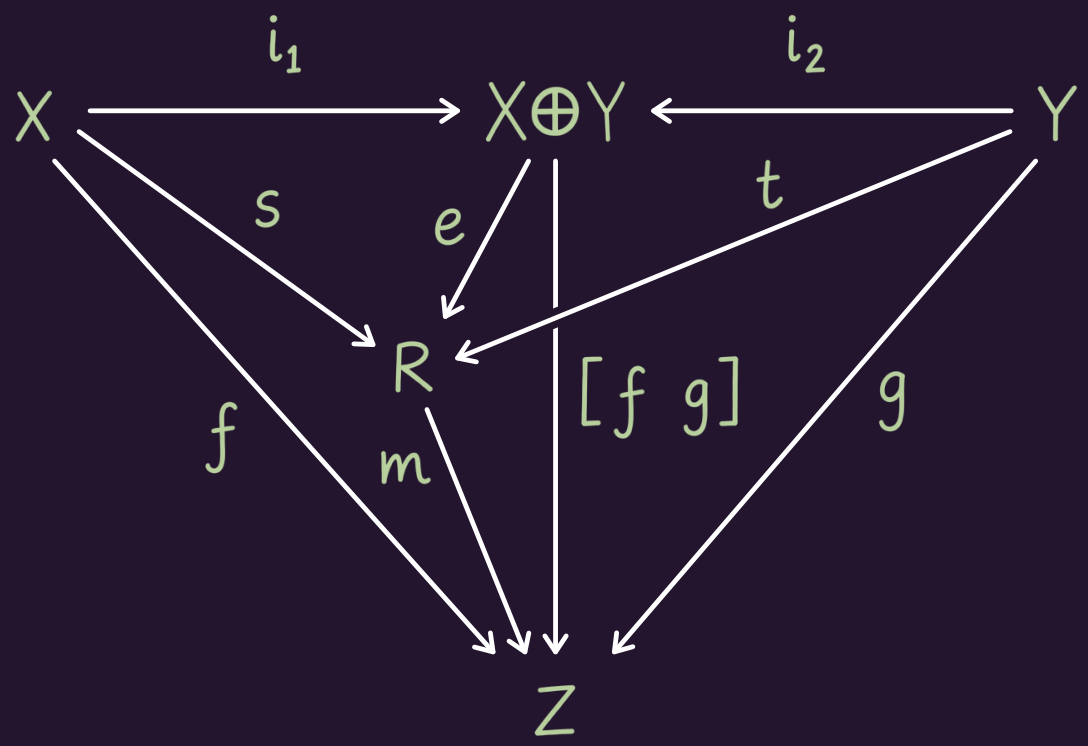
$$\Downarrow$$

$$ss^* = tt^*$$

$$\Downarrow$$

$$1 = ee^* = ss^* + tt^* = 2tt^*$$

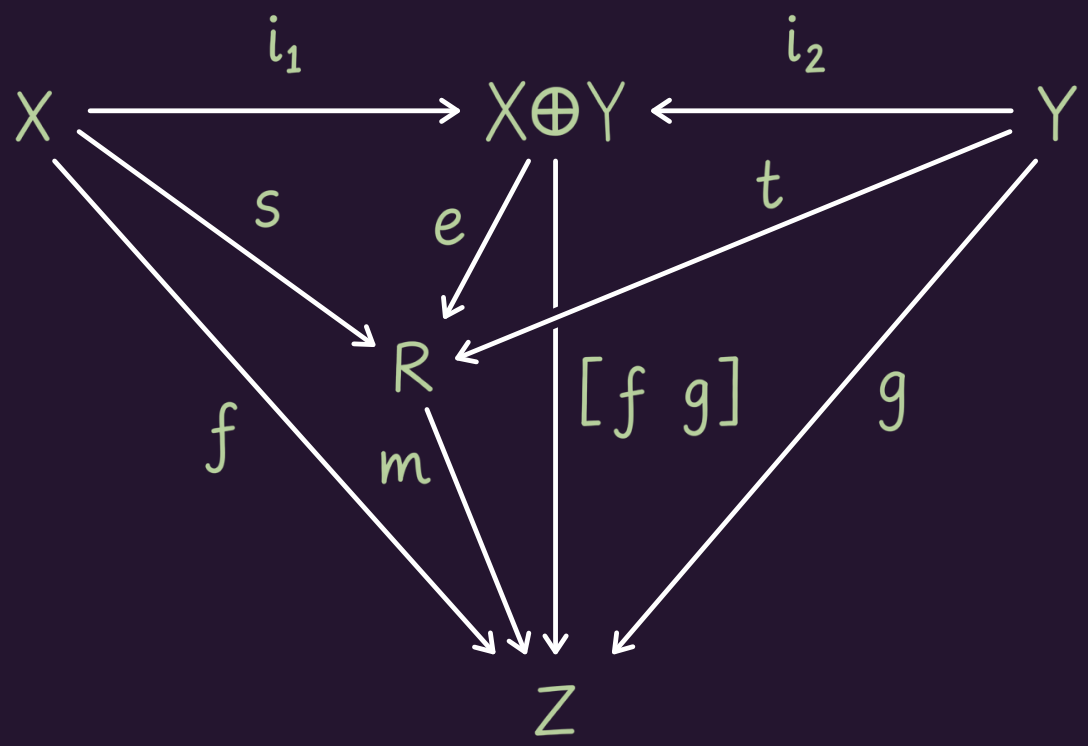
The Algebraic Proof



$h = 2t^*s$

$ff^* = gg^* \Rightarrow f = gh,\ g = fh^*,\ h = hh^*h$

The Algebraic Proof

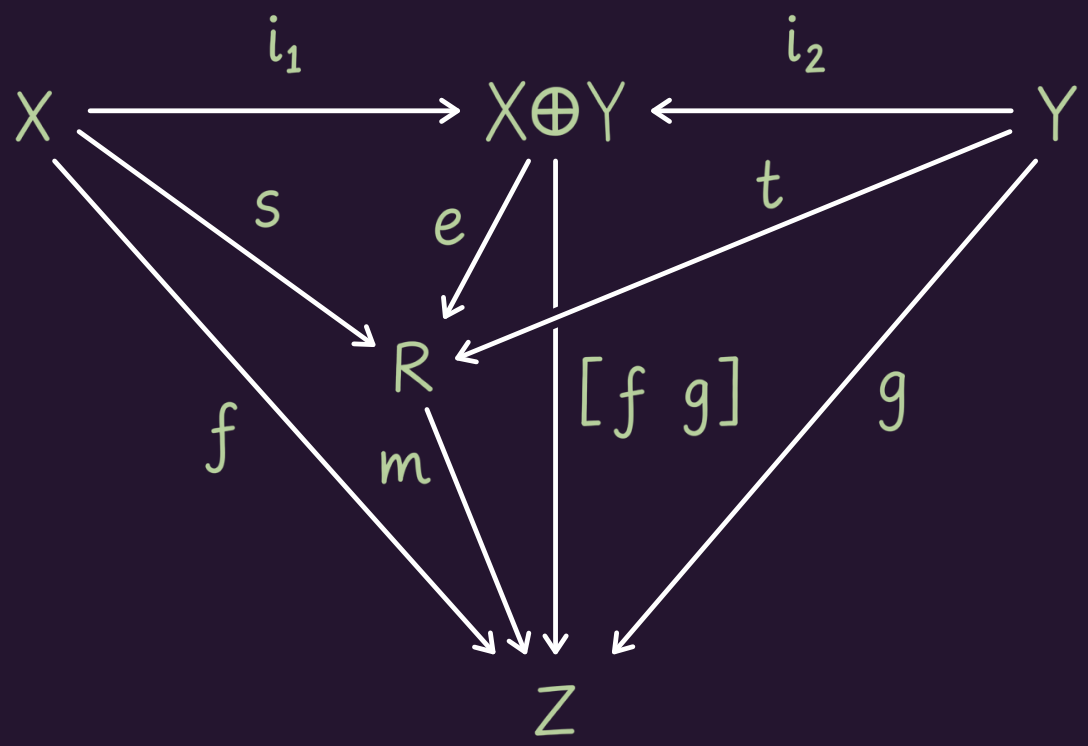


$h = 2t^*s$

$gh = mt(2t^*s)$

$ff^* = gg^* \Rightarrow f = gh,\ g = fh^*,\ h = hh^*h$

The Algebraic Proof



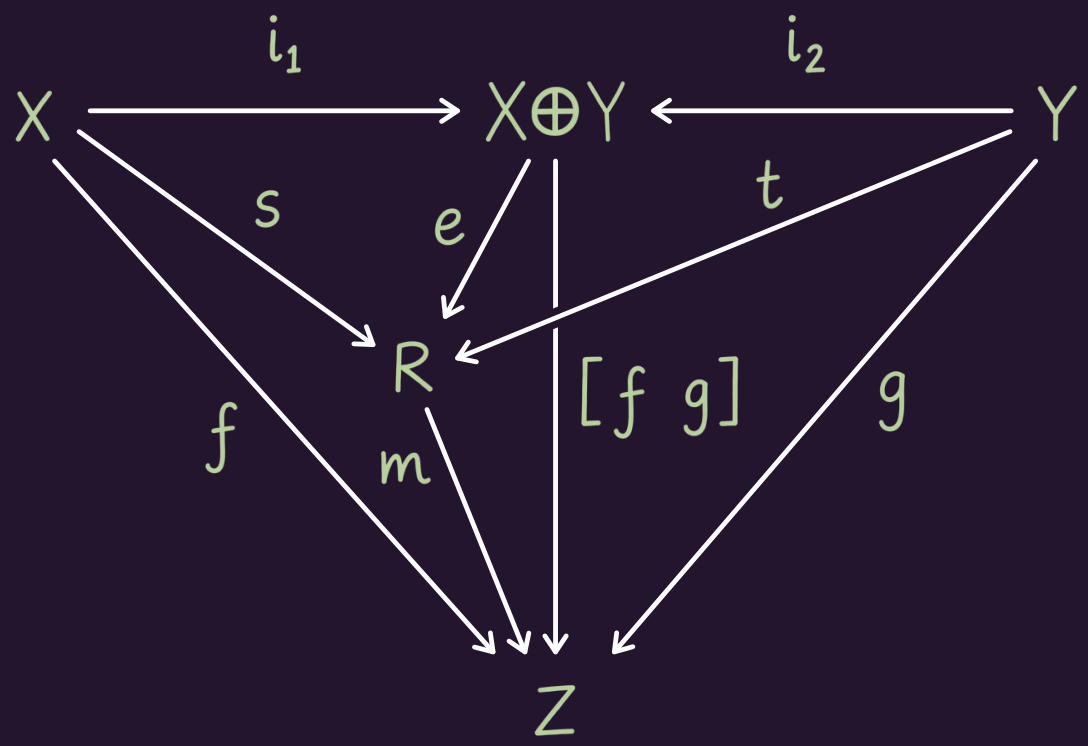
$h = 2t^*s$

$gh = mt(2t^*s)$

$= m(2tt^*)s$

$ff^* = gg^* \Rightarrow f = gh,\ g = fh^*,\ h = hh^*h$

The Algebraic Proof

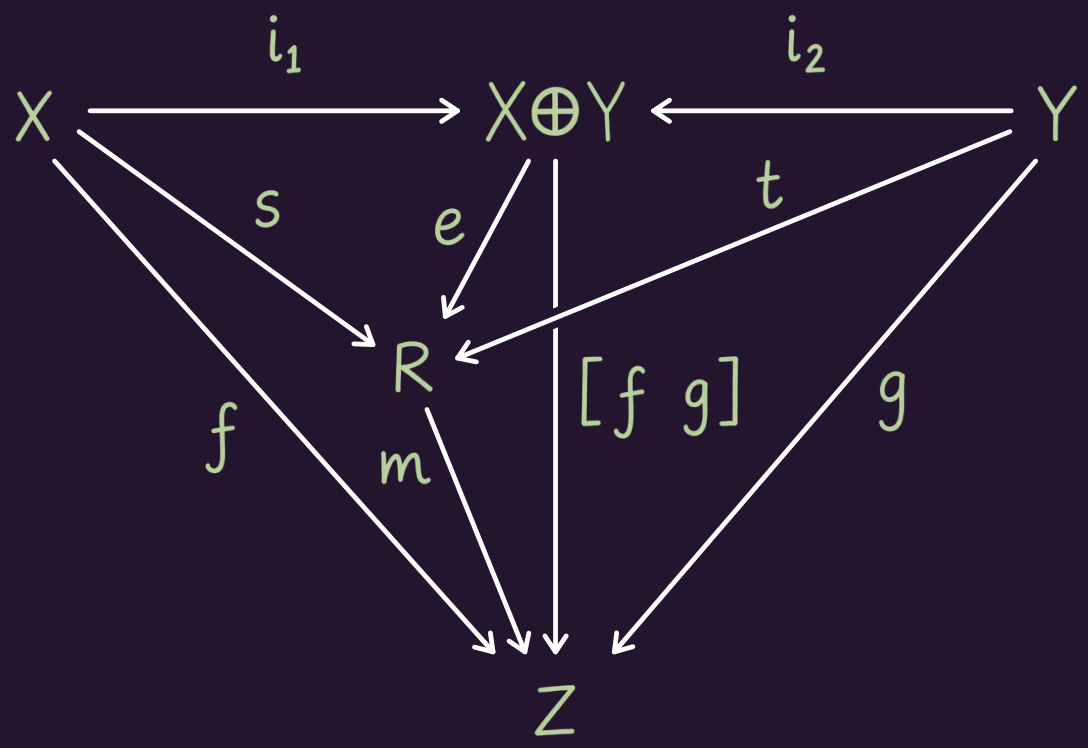


$h = 2t^*s$

$$gh = mt(2t^*s)$$
$$= m(2tt^*)s$$
$$= ms$$

$$ff^* = gg^* \Rightarrow f = gh,\ g = fh^*,\ h = hh^*h$$

The Algebraic Proof

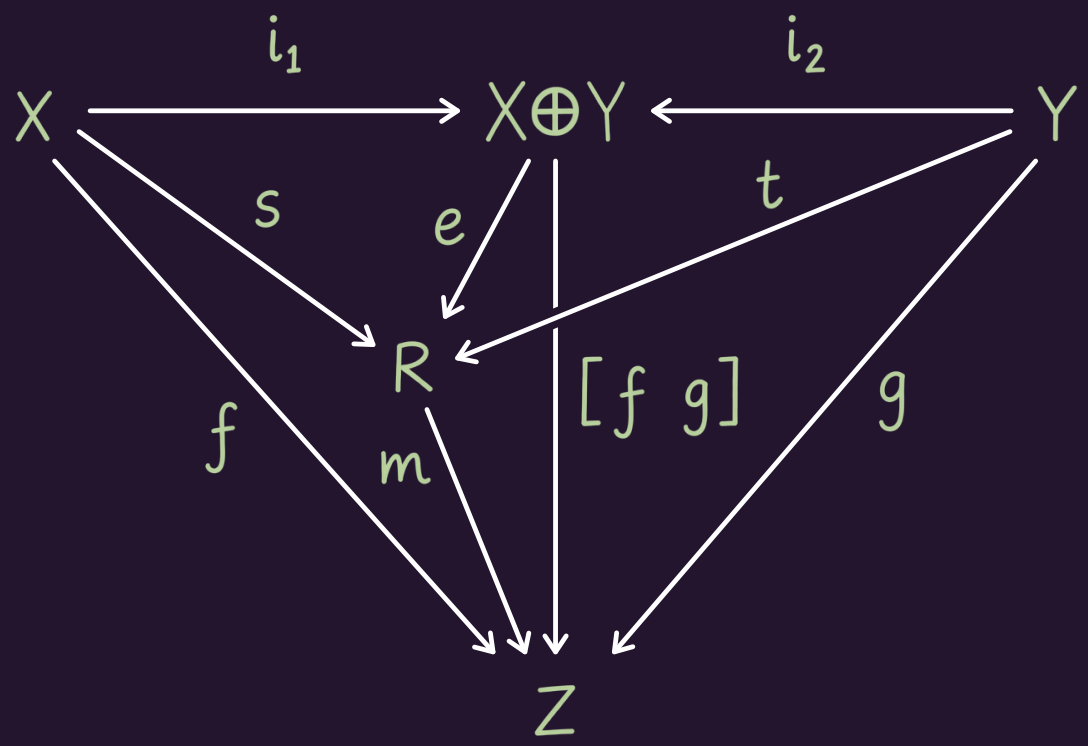


$$ff^* = gg^* \implies f = gh,\ g = fh^*,\ h = hh^*h$$

$$h = 2t^*s$$

$$\begin{aligned} gh &= mt(2t^*s) \\ &= m(2tt^*)s \\ &= ms \\ &= f \end{aligned}$$

The Algebraic Proof



$ff^* = gg^* \Rightarrow f = gh,\ g = fh^*,\ h = hh^*h$

$h = 2t^*s$

$gh = mt(2t^*s)$

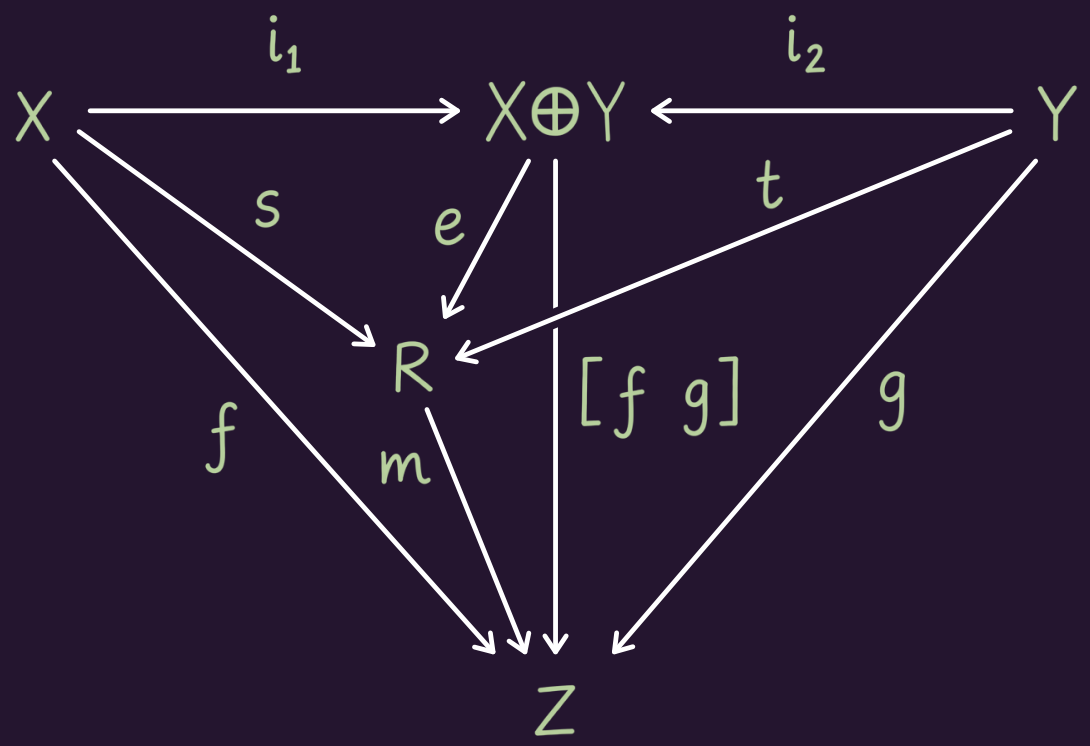
$= m(2tt^*)s$

$= ms$

$= f$

$hh^*h = (2t^*s)(2s^*t)(2t^*s)$

The Algebraic Proof



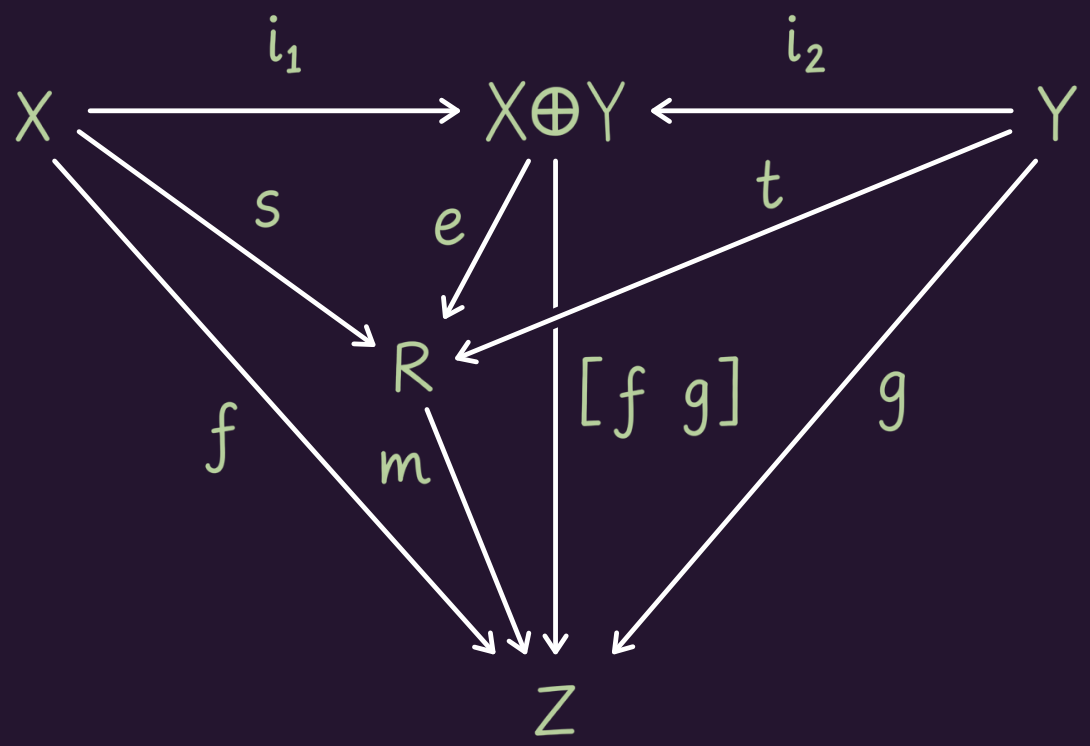
$$ff^* = gg^* \implies f = gh,\ g = fh^*,\ h = hh^*h$$

$$h = 2t^*s$$

$$\begin{aligned} gh &= mt(2t^*s) \\ &= m(2tt^*)s \\ &= ms \\ &= f \end{aligned}$$

$$\begin{aligned} hh^*h &= (2t^*s)(2s^*t)(2t^*s) \\ &= 2t^*(2ss^*)(2tt^*)s \end{aligned}$$

The Algebraic Proof



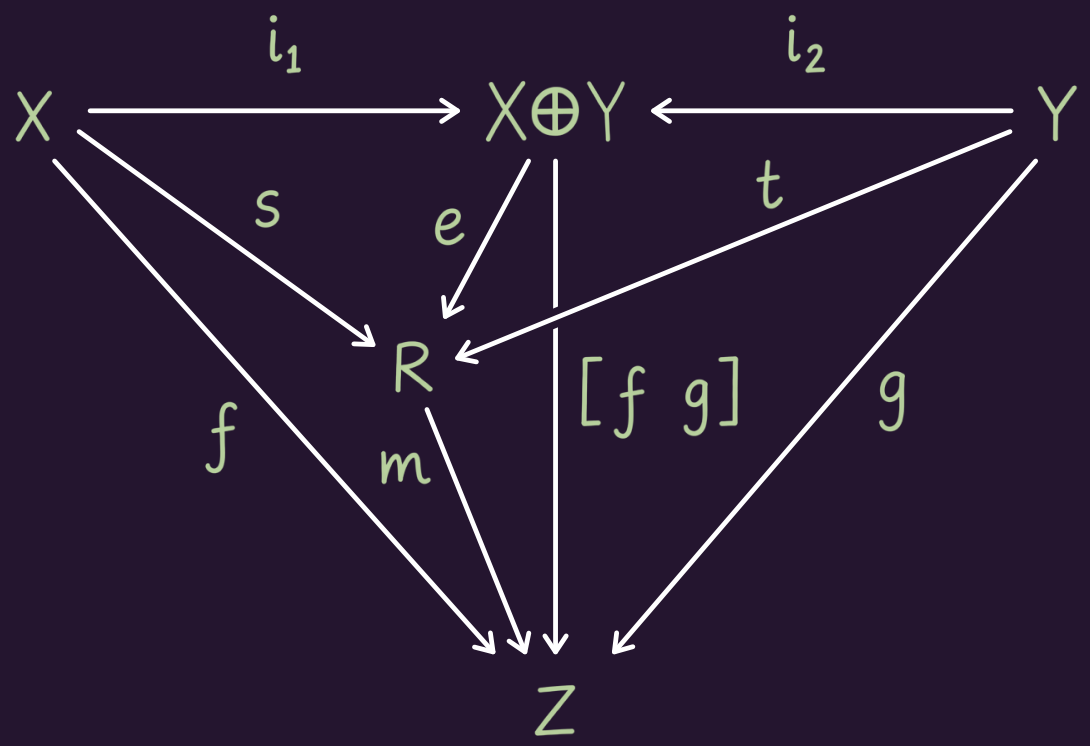
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RECAP

Proved an analytic result using finitary algebraic axioms

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abelian categories : Ab

pre-Hilbert $*$-categories : Hilb

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Proved an analytic result using finitary algebraic axioms

abelian categories : Ab

pre-Hilbert $*$-categories : Hilb

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PRE-HILBERT $*$-CATEGORIES

THE HILBERT-SPACE ANALOGUE OF ABELIAN CATEGORIES

MATTHEW DI MEGLIO

Abstract. This article introduces pre-Hilbert $*$-categories—an abstraction of categories exhibiting "algebraic" aspects of the theory of Hilbert spaces. Notably, finite biproducts in pre-Hilbert $*$-categories can be orthogonalised using the Gram–Schmidt process, and generalised notions of positivity and contraction support variants of Douglas' lemma and Sz.-Nagy's unitary dilation theorem. Underpinning these generalisations is the structure of an involutive identity-on-objects contravariant endofunctor, which encodes adjoints of morphisms. The axioms for pre-Hilbert $*$-categories are otherwise inspired by those for abelian categories, comprising a few simple properties of products and ker-